

30 Days SQL Micro Course

Day 2 – Types of Commands in SQL(DCL)

What is DCL?

DCL (Data Control Language) is used to control user access and permissions in a database.

DCL Commands

Command Purpose

GRANT Give privileges to users

REVOKE Remove privileges from users

1. GRANT Command

The GRANT command is used to give permissions to a user.

Syntax

```
GRANT privilege_name  
ON table_name  
TO user_name;
```

Example 1: Grant SELECT Permission

```
GRANT SELECT  
ON Students  
TO user1;
```

Result: user1 can view data from the Students table.

Example 2: Grant Multiple Permissions

```
GRANT SELECT, INSERT, UPDATE  
ON Students  
TO user1;
```

Result: user1 can:

- View data

- Add records
 - Modify records
-

Example 3: Grant All Permissions

**GRANT ALL PRIVILEGES
ON Students
TO user1;**

Result: user1 gets all available permissions on the table.

2. REVOKE Command

The REVOKE command removes permissions from a user.

Syntax

**REVOKE privilege_name
ON table_name
FROM user_name;**

Example 1: Revoke SELECT Permission

**REVOKE SELECT
ON Students
FROM user1;**

Result: user1 can no longer view data from the table.

Example 2: Revoke Multiple Permissions

**REVOKE INSERT, UPDATE
ON Students
FROM user1;**

Result: user1 loses INSERT and UPDATE permissions.

Example 3: Revoke All Permissions

**REVOKE ALL PRIVILEGES
ON Students
FROM user1;**

Result: All granted permissions are removed.

Complete Example

Grant Permission

**GRANT SELECT, INSERT
ON Students
TO user1;**

User Can

SELECT * FROM Students;

**INSERT INTO Students
VALUES (1, 'Amit', 20);**

Revoke Permission

**REVOKE INSERT
ON Students
FROM user1;**

Now

**INSERT INTO Students
VALUES (2, 'Neha', 21);**

✗ Permission denied

Common Interview Questions

1. What is DCL?

Answer: DCL (Data Control Language) manages user permissions and access rights.

2. Which commands belong to DCL?

Answer:

- **GRANT**
 - **REVOKE**
-

3. What is GRANT used for?

Answer: GRANT gives permissions to a user.

**GRANT SELECT
ON Students
TO user1;**

4. What is REVOKE used for?

Answer: REVOKE removes permissions from a user.

**REVOKE SELECT
ON Students
FROM user1;**

5. Can multiple permissions be granted at once?

Answer: Yes.

**GRANT SELECT, INSERT, UPDATE
ON Students
TO user1;**

6. What is the difference between GRANT and REVOKE?

GRANT	REVOKE
Gives permissions	Removes permissions
Allows access	Denies access

7. What is ALL PRIVILEGES?

Answer: Gives or removes all available permissions.

**GRANT ALL PRIVILEGES
ON Students
TO user1;**

8. Is GRANT a DDL, DML, or DCL command?

Answer: DCL

9. Is REVOKE a DDL, DML, or DCL command?

Answer: DCL

10. Why are DCL commands important?

Answer: They help secure the database by controlling who can access or modify data.

Viva / MCQ Questions

Which command gives permissions?

Answer: GRANT

Which command removes permissions?

Answer: REVOKE

Which SQL category manages user access?

Answer: DCL

Can GRANT be used on tables?

Answer: Yes.

Can REVOKE remove multiple permissions?

Answer: Yes.

Quick Summary

Command Purpose

GRANT Give privileges

REVOKE Remove privileges

Memory Tip

- **GRANT = Give Access** ☒
- **REVOKE = Remove Access** ✕

Example:

GRANT SELECT ON Students TO user1;

REVOKE SELECT ON Students FROM user1;